I^{2}MEP-Net: Inter- and Intra-Modality Enhancing Prototypical Network for Few-Shot Hyperspectral Image Classification

Wenzhen Wang, Fang Liu, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— Prototypical network, celebrated for its flexible network structure and metric computation capability, has become a prevailing strategy in addressing challenges associated with few-shot hyperspectral image classification (HSIC). However, factors, such as insufficient training samples, spectral mixing, and noise interference, in complex scenarios severely impact the stability of its prototypes, ultimately leading to a degradation in classification performance. Therefore, this article proposes a novel method called inter- and intra-modality enhancing prototypical network (I^{2}MEP-Net), which incorporates two auxiliary modalities to facilitate both inter- and intra-modality enhancing prototype learning (Intra-MEPL) with the base modality. Specifically, I^{2}MEP-Net employs the auxiliary LiDAR modality with base HS modality from the same scene for inter-modality enhancing prototype learning (Inter-MEPL), which offsets the sparsity of few-shot features through a cross-modal approach. In addition, it utilizes the target few-shot labeled data as the auxiliary HS modality for Intra-MEPL on the enhanced prototypes, in a way that adaptively generates diversity features, thereby further enriching the prototype embedding space and achieving more fine-grained and stable prototypes. Comprehensive experiments are conducted on the publicly available hyperspectral (HS) image datasets. These experiments indicate that the proposed I^{2}MEP-Net outshines the existing state-of-the-art deep learning techniques and few-shot classification methodologies.**

***Index Terms*— Enhancing prototype learning, hyperspectral image classification (HSIC), inter- and intra-modality, prototypical network.**

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Wenzhen Wang is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: wangwz@njust.edu.cn).

Fang Liu is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, also with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China, and also with Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology, Nanjing 210094, China. (e-mail: liufang_cs@njust.edu.cn).

Liang Xiao is with the School of Computer Science and Engineering and the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: xiaoliang@mail.njust.edu.cn).

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I. Introduction

HYPERSPECTRAL (HS), known for its ability to capture diagnostic spectral details of land cover, uniquely combines spatial and spectral unities. HS image classification (HSIC) serves as one of the critical techniques employed in remote sensing interpretation. This process allocates semantic class labels to each pixel within an image, based on their specific spatial–spectral properties, making it widely applicable.

Historically, HSIC is predominantly driven by conventional machine learning methodologies, such as support vector machines (SVMs) [1], K-nearest neighbors (KNNs) [2], and Bayesian classifiers [3]. However, these methods, being shallow models, heavily rely on human expertise for feature selection, thereby limiting their capacity to effectively capture complex features within the data. The rise in computational prowess and expansion in data volume pave the way for deep learning techniques, such as convolutional neural networks (CNNs) [4], graph convolutional networks (GCNs) [5], transformer models [6], and their various variants [7], [8]. These methodologies harness deep neural networks to autonomously learn more abstract, expressive feature representations from original data, contributing significantly to advancements in HSIC. However, they typically require a substantial volume of labeled data, posing a challenge due to the practical reality of new datasets often having limited or no labeling.

To reduce labeling costs, extensive research is undertaken in the field of few-shot learning [9], [10]. Among these, methods based on meta-learning [11] have garnered widespread attention due to their superior generalizability and flexibility. Notably, these approaches transfer shared meta-knowledge learned across different tasks from seen to unseen classes, circumventing the tedious process of training from scratch. From the perspective of task transfer, this can significantly cut down on time expenditure. Metric-based meta-learning methods, such as prototypical network [12], matching network [13], Siamese network [14], and relation network [15], and nonparametric algorithms are extensively implemented to solve few-shot HSIC problems. In particular, popular is prototypical network, given its flexible network structure and metric computation capability. Such methods strive to learn a nonlinear embedding space, where class affiliation is determined by measuring distance between

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points for inferred similarity. Yet, the existing prototypical network-based HSIC methods still encounter two major issues leading to prototype instability: 1) limited training samples inadequately covering sample diversity and 2) the insufficiency of a unimodal mechanism in handling spectral mixing and noise interference in complex scenarios.

For problem 1), traditional prototypical networks are easily influenced by the quantity and quality of selected support sets and struggle with representative prototype predictions in complex scenarios, as shown in Fig. 1(a). In the field of natural language generation, diversity sample generation strategy based on attribute differences creates new samples relevant to known conditions, enriching the dataset and enhancing sample diversity. As demonstrated in Fig. 1(b) with text modality, for a sample "king" that belongs to the attribute "identity" and typically indicates "man," we can generate a new sample "queen" by changing the gender to "woman" while maintaining its belonging to the "identity" attribute. This process can be regarded as category expansion under the condition of the semantic difference attribute, representing a new data augmentation method that has been successfully applied to enhance prototype under sparse samples [16]. Currently, this method is limited to the text modality, often characterizing diverse attributes based on semantic information. We attempt to extend this approach to more modalities, such as enhancing categories within HS scene images using scene background difference attributes, or within land object spectral data by exploiting spectral intensity difference attributes. However, accurate mining and representation of diversity attributes among HS data are a complex and challenging task.

For problem 2), the presence of spectral mixing and noise interference, such as the spectrum of buildings partially covered by green vegetation and the land object spectrum obtained during spectral collection with cloud and fog obstruction, makes it impossible to accurately identify object categories based solely on this spectral information. This inherent challenge in HS images still exists in prototype learning methods, and limited labeled samples can even exacerbate the problem. Studies [17], [18], [19], [20] show that multimodal data, including text, audio, and so on, can reduce ambiguity in unimodal few-shot learning and improve model generalization in visual image classification tasks. Ma et al. [21] are the first to systematically explore prototype-based multimodal few-shot learning, dividing it into deterministic prototypical networks and shifted prototypical networks. As illustrated in Fig. 1(c) and (d), deterministic prototypical network fuses multimodal data for the joint multimodal prototype representations using a fusion model, while shifted prototypical network integrates prototypes of each modality in a way that cotrains the auxiliary modality with the base modality. Compared with deterministic prototypical networks that focus more on multimodal fusion learning, shifted prototypical networks incorporate additional meta-knowledge during training using auxiliary modalities, enriching and diversifying the embedding space. Zhang et al. [22] successfully introduce a text labeling language model into HSIC, improving visual representation learning. However, text labels only characterize class commonalities and fail to describe individual differences. In complex scenarios, complementary modalities, like LiDAR, SAR, and so on, can provide more comprehensive information to improve HSIC accuracy.

Therefore, to address the two main challenges currently faced by prototypical network-based few-shot HSIC, we propose an innovative few-shot HSIC method called inter- and intra-modality enhancing prototypical network (I^2MEP-Net). This method introduces two auxiliary modalities individually for inter- and intra-modality enhancing prototype learning (Intra-MEPL) with the base modality. On the one hand, we involve auxiliary LiDAR modality with base HS modality from the same scene to colearn multiple embedding prototypes, refining prototype representations via a weighted average technique. On the other hand, aiming to maintain the intrinsic diversity of the base modality in the refined prototypes, we further use the target few-shot data as the auxiliary HS modality for diversity feature generation. This process amplifies individual features of prototypes, consequently enriching the prototype embedding space. The main contributions of this work can be summarized as follows.

1) A novel method called I^2MEP-Net is proposed in this article for few-shot HSIC. It improves the few-shot prototype representations in complex scenarios by exploiting the prior knowledge provided by auxiliary modalities.
2) The proposed method introduces the Inter- and Intra-MEPL module. It employs the auxiliary LiDAR modality from the same scene to cooperate with the base HS modality for inter-modality enhancing prototype learning (Inter-MEPL) and then uses the auxiliary HS modality from the target scene to perform Intra-MEPL, and ultimately, the more fine-grained and stable prototypes can be obtained.
3) The I^2MEP-Net expands beyond the scope of multimodal enhancing prototype learning for visual and textual modalities only, and it outperforms the state-of-the-art methods on the publicly HSIC datasets, which provides a new research idea for multimodal few-shot learning.

The rest of this article is organized as follows. Section II illustrates the related work. The proposed method in detail is described in Section III. Section IV shows the experimental results, and the discussion of details is given in Section V. Finally, conclusion is given in Section VI.

II. Related Work

A. Prototypical Network

For prototypical network-based few-shot classification, a large-scale source dataset $\mathbf{X}^S_{\text{base}}$ contains many-shot labeled data from each base class in C^S, and a small-scale target dataset $\mathbf{X}^{\mathcal{T}}_{\text{novel}}$ consists of few-shot labeled data from each novel class in $C^{\mathcal{T}}$, as well as a certain number of unlabeled data acquired also from the category set $C^{\mathcal{T}}$. Note that, $C^S \gg C^{\mathcal{T}}$. For an $N-$way $K-$shot $Q-$query problem ($N = C^{\mathcal{T}}$), we randomly select N classes from dataset $\mathcal{D}^S_{\text{base}}$ and choose K labeled samples per class to calculate the prototype vectors for each class. These samples are collectively referred to as the support set $S = \{x_s, y_s\}_{s=1}^{N\times K}$, while the remaining data are

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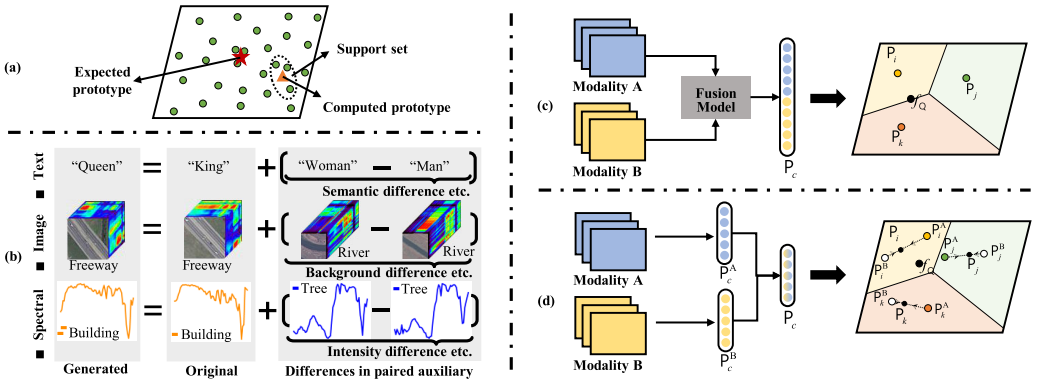


Fig. 1. Illustrations for the instability issue of traditional prototype learning. Left: illustrations show the issue and one possible solution. (a) Illustration of the bias between the computed prototype and the expected prototype in the traditional prototypical network. The computed prototype is obtained by taking the mean of the support set. (b) Illustration of sample generation based on diversity representation. Difference in paired auxiliary samples is used as diversity representation to guide new sample generation from the same class in some form. Right: flow of multimodal prototype learning. (c) Deterministic prototypical network. It fuses multimodal data for the joint multimodal prototype representations using a fusion model. (d) Shifted prototypical network. It integrates prototypes of each modality in a way that cotrains the auxiliary modality with the base modality.

used to form a query set $Q = \{x_q, y_q\}_{q=1}^{N\times Q}$, consisting of Q samples per class. Following the task-trained learning manner of meta-learning, each task consists of a support set S and a query set Q, and a prototypical network $\mathcal{F}_\theta$ is utilized to extract embedding features from the data. Subsequently, the average embedding feature of the support set is employed to compute the embedding prototype P_c for each class c

$$P_c = \frac{1}{|S_c|}\sum_{(x_s,y_s)\in S_c}\mathcal{F}_\theta(x_s) \tag{1}$$

where $S_c \subset S$ is a subset of the support set belonging to the category c, $|S_c| = K$.

Then, the class affiliation is predicted based on the softmax value of the similarity distance, computed from the embedding features of the query data x_q to the embedding prototypes P_c

$$p_\theta\left(y_q = c\middle|x_q\right) = \frac{\exp\left(-d\left(\mathcal{F}_\theta\left(x_q\right), P_c\right)\right)}{\sum_{c=1}^{N}\exp\left(-d\left(\mathcal{F}_\theta\left(x_q\right), P_c\right)\right)} \tag{2}$$

where y_q is the label of x_q and d denotes the Euclidean distance function.

The objective of the network is to learn a prototype that can acquire general knowledge and generalization ability based on thousands of tasks. The classification loss function for each task can be defined as the negative log probability of the true class c

$$\mathcal{L}_{\text{fsl}} = \mathbb{E}_{S,Q}\left[-\sum_{(x_q,y_q)\in Q}\log p_\theta\left(y_q = c\middle|x_q\right)\right]. \tag{3}$$

The network is trained by minimizing this loss, and finally, a trained prototypical network with N classification capabilities is used to encode the target data. In the few-shot HSIC tasks, a nearest neighbor classifier (NN Classifier) is typically learned from the encoded features of target few-shot labeled data, which is then used to classify the target unlabeled data $\mathsf{U} \subset \mathsf{X}_{\text{novel}}^{\mathcal{T}}$.

B. Enhancing Prototype Learning

Enhanced prototype learning is an effective way to address the lack of representativeness in prototype centers caused by sparse samples. Liu et al. [23] propose cross-enhanced prototypes that acquire more universal prototypes by reducing intraclass variations. Yu et al. [24] initiate the transformation of prototypes gained from average vectors into local HS image patch aggregates using a calibration aggregation network. Recent research leans toward the most direct and effective ways to enhance data scale utilization and reduce the bias between samples and prototypes, such as methods leveraging unlabeled query data to extend the support set for more representative class prototype acquisition [25], [26], [27], [28]. However, this method is overly reliant on the distribution and features of query samples, which may lead to a lack of diversity and generalizability in the new prototypes. Yang et al. [16] propose a diversity semantic-enhanced prototypical network that generates samples based on intraclass variation diversity representation in text, enriching the feature space of the support set. The method combines two techniques of prototype learning and data augmentation, which eliminates the dependence on external data, such as query sets in general methods, and generates diversity samples that are more helpful to improve the model's ability to model complex data distributions. But, for complex HS data, how to represent diversity is one of the challenging issues in such methods.

Although the aforementioned methods have improved prototype representation to some extent through enhanced prototype learning, they mainly target single-type data for processing and modeling, lacking exploration for multimodal data.

C. Shifted Multimodal Prototype Learning

Shifted multimodal prototype learning enhances the generalization ability in new tasks of meta-learners by integrating the advantages of different modalities. Xing et al. [29] adaptively mixed category labeled prototypes in visual modality to alleviate prototype bias under sparse data conditions. The computation form of mixed prototype is

$$P_c' = \lambda * P_c + (1-\lambda) * P_c^{\text{t}} \tag{4}$$

where P_c^{t} represents the prototype representations of category labels in the same embedding space and λ is an adaptive weight parameter. Schwartz et al. [30] and Pahde et al. [31]

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add multiple semantic labels in addition to using category labels, improving the computation of auxiliary prototypes. Specifically, in [31], a generative adversarial learning method is used to align the auxiliary modality with the base modality, compensating for the sparseness of support set samples. Its multimodal prototype calculation formula is

$$P'_c = \frac{P_c + \lambda * P_c^{\mathrm{t}}}{1+\lambda} \tag{5}$$

where $P_c^{\mathrm{t}} = (1/|S_c^{\mathrm{t}}|) \sum_{(x_s^{\mathrm{t}}, y_s^{\mathrm{t}}) \in S_c^{\mathrm{t}}} G(t(x_s^{\mathrm{t}}))$, G is the generator, and t is the text encoder. Furthermore, Mu et al. [32] propose language-shaped learning, which encourages the embedding space of learned visual prototypes to decode sample category information. It could be said that shifted multimodal prototype learning enhances prototype representations in low data environments by flexibly representing and combining various modalities. In addition, simple metric computation methods, such as Euclidean distance, cosine distance, contrast loss, and triplet loss, provide convenience for shifted multimodal prototype learning.

However, most current research confines modalities to visual and textual modalities or pairwise occurring dual modalities, which poses challenges for prototype learning frameworks that are more modalities.

III. Proposed Method

The few-shot HSIC task is redefined in our work. A large-scale multimodal source dataset $\mathsf{X}_{\mathrm{base}}^{S}$ contains many-shot labeled HS modality data and its corresponding LiDAR modality data from each base class in C^S. The labels for these two modalities are known and shared. Also, a small-scale unimodal target dataset $\mathsf{X}_{\mathrm{novel}}^{\mathcal{T}}$ consists of few-shot labeled HS modality data from each novel class in $C^{\mathcal{T}}$, as well as a certain number of unlabeled data acquired also from the category set $C^{\mathcal{T}}$. Note that, $C^S \gg C^{\mathcal{T}}$; to evaluate the method, unlabeled data and their corresponding labels in the target dataset exist but do not participate in training. Our goal is to learn an enhanced prototype embedding space using multimodal data for unimodal unlabeled data classification of target tasks. In particular, the I^{2}MEP-Net is proposed, as shown in Fig. 2, and consists of meta-task construction, feature extraction, and Inter- and Intra-MEPL. It innovatively uses multimodal source data to help unimodal target data for cross-modal and cross-domain few-shot learning.

A. Meta-Task Construction

Different from classic unimodal prototype learning, our proposed I^{2}MEP-Net utilizes two auxiliary modalities to collaborate with the base modality to conduct Inter- and Intra-MEPL. In the proposed few-shot HSIC task, the HS modality data in $\mathsf{X}_{\mathrm{base}}^{S}$ are taken as the base modality $\mathsf{X} \in \mathbb{R}^{HW \times B}$ for initial prototype learning, while the corresponding LiDAR modality data $\mathsf{X}^{\mathcal{A}_1} \in \mathbb{R}^{HW \times B_1}$ are used to assist the base prototype for Inter-MEPL. Notably, we need to be rigorous in requiring category alignment between the auxiliary LiDAR data and the base HS data, which is key to learning a common prototype embedding space. For Intra-MEPL, the auxiliary HS data need to provide as much spectral difference information as possible to ensure the validity of diversity representation. In addition, it has been proved in [33] that incorporating target few-shot labeled data into the learning process helps classifiers more robustly apply to unknown data classification in the target datasets. Therefore, the target few-shot labeled HS data in $\mathsf{X}_{\mathrm{novel}}^{\mathcal{T}}$ are served as the auxiliary HS modality $\mathsf{X}^{A_2} \in \mathbb{R}^{Nl \times B_2}$, where Nl denotes the number of target few-shot labeled HS data.

Then, X, X^{A_1}, and X^{A_2} are sampled in cube forms of s size, respectively, and sample sets $\{x_1, x_2, x_3, \ldots\} \in \mathbb{R}^{s \times s \times b}$, $\{x_1^{A_1}, x_2^{A_1}, x_3^{A_1}, \ldots\} \in \mathbb{R}^{s \times s \times B_1}$, and $\{x_1^{A_2}, x_2^{A_2}, x_3^{A_2}, \ldots\} \in \mathbb{R}^{s \times s \times b}$ are obtained, where b represents the uniform number of HS modality channel mappings obtained by 1×1 convolution. Therefore, for an $N-$ way $K-$ shot $Q-$ query problem, the constructed meta-task includes a support set $S = \{x_s, y_s\}_{s=1}^{N \times K}$ and a query set $Q = \{x_{\mathrm{q}}, y_{\mathrm{q}}\}_{\mathrm{q}=1}^{N \times Q}$ from X, an auxiliary set $A_1 = \{x_i^{A_1}, y_i^{A_1}\}_{i=1}^{N \times K}$ from X^{A_1}, and an auxiliary set $A_2 = \{x_{i,1}^{A_2}, x_{i,2}^{A_2}, y_i^{A_2}\}_{i=1}^{N}$ from X^{A_2}, where $x_{i,1}^{A_2}$ and $x_{i,2}^{A_2}$ are a randomly sampled pair of labeled samples that share the label $y_i^{A_2}$, used for the diversity representations.

B. Feature Extraction

Given that data from different modalities bear distinct characteristics, dimensions, and expressive capabilities, it is necessary to design corresponding feature encoders for each modality to obtain more representative and distinctive feature representations.

1) HS Modality: A 3-D residual network can effectively handle high-dimensional HS data, capturing correlations among spectral, spatial, and depth information. Through deep residual learning, the model's capability in modeling HS data can be enhanced. In our proposed approach, we employ a two-layer 3-D residual CNN network $\mathcal{F}_\theta$ to encode HS modality feature. This encoder, parameterized by θ, learns a d_1-dimensional space. Within this space, samples belonging to the same category are closer together, while samples of different categories are farther apart. Therefore, $\mathcal{F}_\theta$ serves as a prototypical network within our framework that is used to encode features for both the base HS modality and the auxiliary HS modality, thus enhancing the generalization capability to the target data. In particular, the encoded features of support set S can be directly computed to obtain the initial prototype P_c of per class c, i.e.,

$$P_c^{(0)} = \frac{1}{|S_c|} \sum_{(x_s, y_s) \in S_c} \mathcal{F}_\theta(x_s) \tag{6}$$

where $S_c \subset S$ is a subset of the support set belonging to category c. Under the settings $K = 1$ in this article, $|S_c| = 1$.

2) LiDAR Modality: A two-layer 2-D residual CNN network $\mathcal{G}_\varphi$ is employed for LiDAR modality feature encoding in our proposed approach, where $\mathcal{G}_\varphi$ acts as an encoder, parameterized by φ, to learn a d_2-dimensional space. For each class c in the auxiliary set A_1, its encoded feature is

$$g_c^{A_1} = \frac{1}{|A_c^1|} \sum_{\left(x_i^{A_1}, y_i^{A_1}\right) \in A_c^1} \mathcal{G}_\varphi\left(x_i^{A_1}\right) \tag{7}$$

where $A_c^1 \subset A_1$, $|A_c^1| = K = 1$ in this article.

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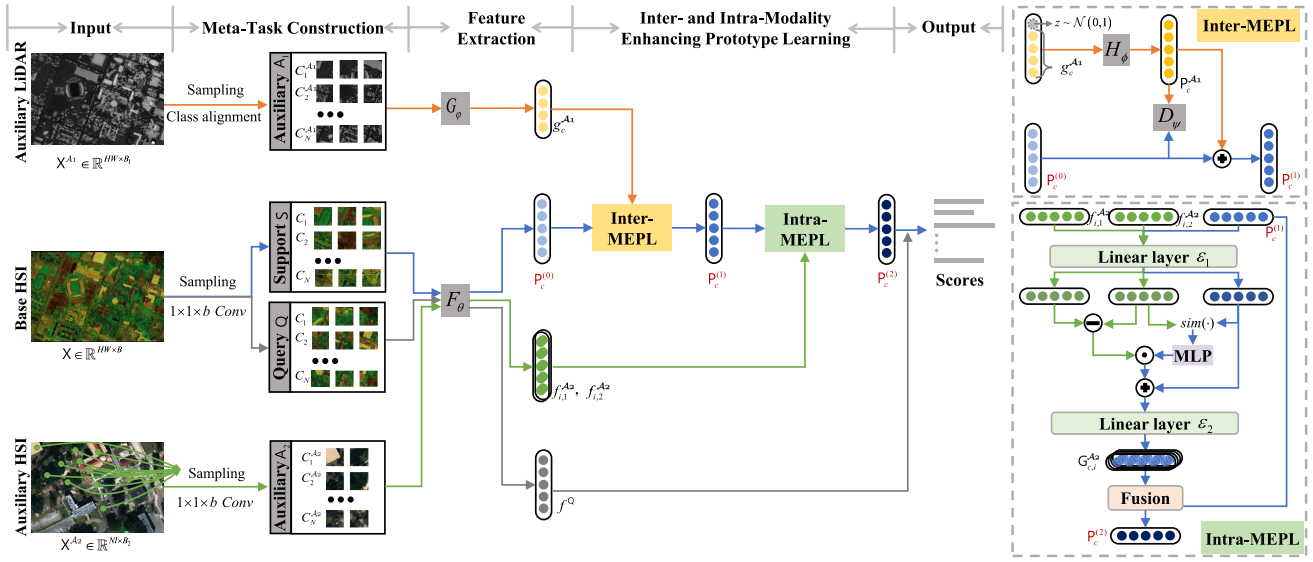


Fig. 2. Proposed I²MEP-Net. It consists of three main modules: meta-task construction, feature extraction, and Inter- and Intra-MEPL. Of these, the designed Inter- and Intra-MEPL module contains two parts: Inter-MEPL and Intra-MEPL, where it utilizes two auxiliary modalities to synergize with the base modality for enhancing prototype learning.

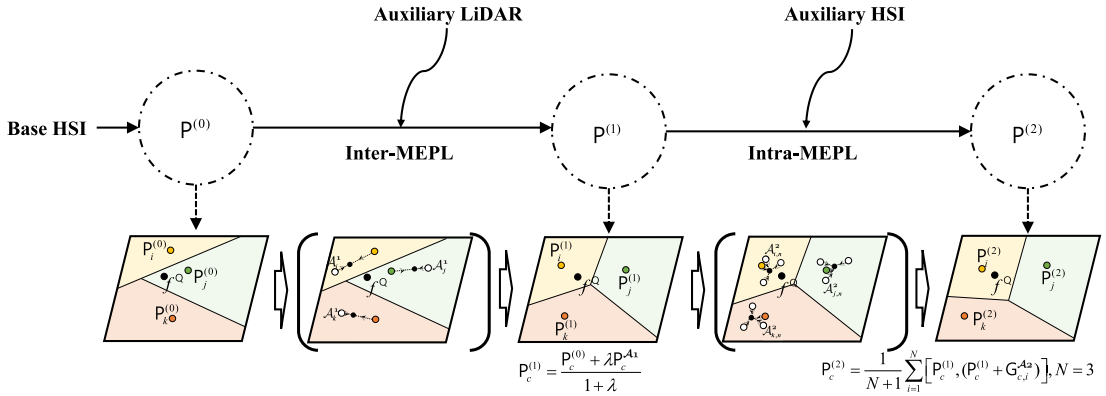


Fig. 3. Qualitative example of how I²MEP-Net works. $P^{(0)}$ denotes the initial prototypes obtained from the base HS modality, $P^{(1)}$ indicates the enhanced prototypes obtained by adding the auxiliary LiDAR modality for Inter-MEPL, and $P^{(2)}$ refers to the stabilized and refined prototypes after adding the auxiliary HS modality for further Intra-MEPL on the basis of the enhanced prototypes and is used as the final prototypes representation. It can be observed that the enhanced prototype embedding space is gradually optimized, which is more conducive to the category prediction of query samples.

C. Inter- and Intra-MEPL

The proposed method leverages prior knowledge provided by auxiliary data from two different modalities for Inter- and Intra-MEPL. As shown in Fig. 3, it first uses the auxiliary LiDAR modality to cooperate with the base HS modality for Inter-MEPL, to compensate for the sparsity of few-shot features. Then, it uses the auxiliary HS modality to perform Intra-MEPL on the enhanced prototypes, to further enrich the prototype embedding space with the adaptive diversity of the feature generation approach, and ultimately obtain more refined and stable prototypes.

1) Inter-MEPL: HS data have high spectral dimension and large processing complexity, making the model prone to overfit with limited labeled samples. Moreover, HS data exhibit weak discrimination of similar material surfaces. So, we utilize auxiliary LiDAR modality data, which has relatively lower computational complexity and high-precision spatial information, to collaboratively conduct Inter-MEPL with the base HS modality data. The Inter-MEPL takes encoded features $g_c^{A_1}$ of class c from auxiliary LiDAR data and initial prototype $P_c^{(0)}$ as inputs. It designs a cross-modal learning method based on generative adversarial networks (GANs) with random noise, generating additional classification features using provided LiDAR information to compensate for the sparsity of few-shot features, hence obtaining enhanced prototype $P_c^{(1)}$. Specifically, the random noise $z \in \mathbb{R}^{d_1-d_2}$, obeying normal distribution, is concatenated to the auxiliary feature and fed into the transformation network $\mathcal{H}_\phi$ for feature dimension conversion, as well as to learn mapping from auxiliary feature to prototype space, resulting in transformed auxiliary feature

$$h_c^{A_1} = \mathcal{H}_\phi\left(\left[g_c^{A_1}, z\right]\right) \tag{8}$$

where the transformation network comprises a 1×1 convolution layer and a normalization layer, ϕ is the network parameter, and $[\cdot, \cdot]$ represents concatenation operation. At this point, auxiliary features from the LiDAR modality are considered as generated extra-classification features, allowing the computation of an auxiliary prototype $P_c^{A_1}$ for per class c

$$P_c^{A_1} = \frac{1}{|A_c^1|} \sum_{\left(x_i^{A_1}, y_i^{A_1}\right) \in A_c^1} \mathcal{H}_\phi\left(\left[\mathcal{G}_\varphi\left(x_i^{A_1}\right), z\right]\right). \tag{9}$$

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The auxiliary prototype enriches the sample space of the auxiliary LiDAR modality in the form of random noise. To better learn cross-modal information provided by the auxiliary LiDAR modality, we adopt a discriminator $\mathcal{D}_\psi$ with parameters ψ for aligning the distribution of auxiliary prototype $P_c^{A_1}$ and initial prototype $P_c^{(0)}$. The feature encoder and transformation network of the LiDAR modality jointly act as the generator, generating auxiliary features indistinguishable from initial prototype, while the discriminator can correctly distinguish between generated auxiliary prototype and initial prototype. Based on such adversarial gameplay, our adversarial learning can be represented as follows:

$$\min_{\varphi,\phi}\max_{\psi}\left(\mathbb{E}_{P_c^{(0)}}\log\left[\mathcal{D}_\psi\left(P_c^{(0)}\right)\right]+\mathbb{E}_{P_c^{A_1}}\log\left[1-\mathcal{D}_\psi\left(P_c^{A_1}\right)\right]\right) \tag{10}$$

where $P_c^{(0)}\sim P_{\mathsf{X}}$ and $P_c^{A_1}\sim P_{\mathsf{X}^{A_1},z}$. P_{X} is the distribution space of dataset X. $P_{\mathsf{X}^{A_1},z}$ denotes the feature distribution of $P_c^{A_1}$, which is influenced by both the data X^{A_1} and the noise z. Thus, the data distribution of $P_c^{A_1}$ is neither fully subject to $P_{\mathsf{X}^{A_1}}$ nor to P_z, but is influenced by the interaction of both, potentially exhibiting a distribution characteristic that lies between the two. Moreover, we add an auxiliary task of class prediction to constrain the generated feature, i.e.,

$$\mathcal{L}_c^{A_1}=\mathbb{E}_{y_i^{A_1},x_i^{A_1}}\log p_{\varphi,\phi}\left(y_i^{A_1}|x_i^{A_1}\right). \tag{11}$$

The losses $\mathcal{L}_d$ and $\mathcal{L}_c^{A_1}$ jointly constrain the auxiliary prototype $P_c^{A_1}$. Then, we can compute the enhanced prototype $P_c^{(1)}$ for per class using the weighted average of two prototypes

$$P_c^{(1)}=\frac{P_c^{(0)}+\lambda P_c^{A_1}}{1+\lambda}. \tag{12}$$

2) Intra-MEPL: Inter-MEPL aligns the initial prototype in sparse-sample scenarios with auxiliary prototype in rich sample space, compensating for the sparsity of few-shot features. Despite this, it might cause the original sparse samples to lose their diversity. Therefore, we need to further enhance the prototype, emphasizing individual characteristics of samples. Data augmentation, one of the simplest means, enriches sample diversity representation by fabricating new samples. However, it cannot guarantee that the new samples will still reflect the core characteristics of the original samples. Our method considers internal feature differences among samples from the same class as diversity feature representations. By adding diversity features to known sample features, sample feature diversity is enriched. This approach preserves the original data information while enhancing the feature space of samples. In addition, the inclusion of target information in the training process helps to improve the generalization ability of the model in the target task. Consequently, the proposed Intra-MEPL uses the target few-shot labeled data from the auxiliary HS modality as the source of diversity feature representation and acquires a more refined and stable prototype in an adaptive diversity feature generation manner.

The Intra-MEPL illustration in Fig. 2 shows the process of performing Intra-MEPL on a prototype $P_c^{(1)}$ of class c, using the ith pair of labeled sample features $\{f_{i,1}^{A_2}, f_{i,2}^{A_2}\}$ from the auxiliary set A_2, where $f_{i,\cdot}^{A_2}=\mathcal{F}_\theta(x_{i,\cdot}^{A_2})$. Initially, they are sent together into a linear layer to obtain feature representations in the same space, and then, the feature differences of auxiliary feature pairs are calculated to represent the diversity features within the same category

$$\Delta f=\varepsilon_1\left(f_{i,1}^{A_2}\right)-\varepsilon_1\left(f_{i,2}^{A_2}\right) \tag{13}$$

where ε_1 and ε_2 represent the linear layers.

To diminish the negative impact caused by irrelevant information in diversity features on the enhanced data, on the one hand, we adopt an attention mechanism to encourage the generation of valid sample features, generating features as follows:

$$\mathsf{G}_{c,i}^{A_2}=\varepsilon_2\left(\omega\odot\Delta f+\varepsilon_1\left(P_c^{(1)}\right)\right) \tag{14}$$

where $\odot$ indicates elementwise multiplication operation and $\omega\in\mathbb{R}^{160}$ is an attention weight learned by MLP, which is computed by the relevance between auxiliary features and the prototype to be enhanced, i.e.,

$$\omega=\mathrm{MLP}\left(\mathrm{sim}\left(\tilde{f},\varepsilon_1\left(P_c^{(1)}\right)\right)\right). \tag{15}$$

If $\mathrm{dis}(\varepsilon_1(P_c^{(1)}),\varepsilon_1(f_{i,1}^{A_2}))\geq\mathrm{dis}(\varepsilon_1(P_c^{(1)}),\varepsilon_1(f_{i,2}^{A_2}))$, $\tilde{f}=\varepsilon_1(f_{i,1}^{A_2})$; else $\tilde{f}=\varepsilon_1(f_{i,2}^{A_2})$. MLP consists of a linear layer, an activation layer, another linear layer, and a sigmoid layer, which additionally with the previously mentioned ε_1 and ε_2 are centrally denoted as $\mathcal{T}_\vartheta$, and ϑ is the learnable parameter. sim represents similarity calculation function, and dis adopts subtraction operation in our work.

Furthermore, we similarly add the auxiliary task of class prediction to constrain the model, and in particular, we impose a class prediction loss on generated sample features to ensure that the generated data belong to the expected category, i.e.,

$$\begin{aligned}\mathcal{L}_c^{A_2}=&\,\mathbb{E}_{y_i^{A_2},x_{i,\cdot}^{A_2}}\log p_\theta\left(y_i^{A_2}|x_{i,\cdot}^{A_2}\right)\\&+\mathbb{E}_{y_{c,i},\mathsf{G}_{c,i}^{A_2}}\log p_{\theta,\varphi,\phi,\vartheta}\left(y_{c,i}=c|\mathsf{G}_{c,i}^{A_2}\right).\end{aligned} \tag{16}$$

The first term in (16) constrains the ability of the prototype to characterize the target data, while the second term ensures the effectiveness of the generated features.

N auxiliary feature pairs can generate N groups of diversity features; hence, for each prototype $P_c^{(1)}$ of class c, N new same-category sample features can be generated. In particular, to maintain the initial characteristics of the generated features, a residual operation is used to further optimize the generated sample features. Ultimately, by concatenating and fusing the enhanced prototype features and the generated sample features, the final enhanced prototype of class c is obtained

$$P_c^{(2)}=\frac{1}{N+1}\sum_{i=1}^{N}\left[P_c^{(1)},\left(P_c^{(1)}+\mathsf{G}_{c,i}^{A_2}\right)\right]. \tag{17}$$

D. Adversarial Learning for Few-Shot Classification

The proposed I^{2}MEP-Net involves datasets from different sources and modalities. In (10), we have aligned the modality features, and furthermore, to mitigate the negative impact caused by the distribution difference of data, we share the aforementioned discriminator $\mathcal{D}_\psi$ to perform adversarial learning on the auxiliary modalities and the base modality,

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respectively. Thus, the discrimination loss for learning modal-invariant and domain-invariant embedding features can be formulated as follows:

$$\begin{aligned}\mathcal{L}_d &= \mathcal{L}_{\mathrm{MA}} + \mathcal{L}_{\mathrm{DA}} \\ &= \underbrace{\left(-\mathbb{E}_S \log[\mathcal{D}_\psi(\mathcal{F}_\theta(S))] - \mathbb{E}_{A_1} \log[1-\mathcal{D}_\psi(\mathcal{H}_\phi([\mathcal{G}_\varphi(A_1), z]))]\right)}_{\text{Modal Adversarial Learning Loss}} \\ &\quad + \underbrace{\left(-\mathbb{E}_S \log[\mathcal{D}_\psi(\mathcal{F}_\theta(S))] - \mathbb{E}_{A_2} \log[1-\mathcal{D}_\psi(\mathcal{F}_\theta(A_2))]\right)}_{\text{Domain Adversarial Learning Loss}}\end{aligned} \tag{18}$$

where $S \sim P_{\mathsf{X}}$, $A_1 \sim P_{\mathsf{X}^{A_1}}$, and $A_2 \sim P_{\mathsf{X}^{A_2}}$. P_{X}, $P_{\mathsf{X}^{A_1}}$, and $P_{\mathsf{X}^{A_2}}$, respectively, represent the data distribution spaces of datasets X, X^{A_1}, and X^{A_2}. $\mathcal{L}_{\mathrm{MA}}$ signifies the modal discrimination loss existing between the base HS and auxiliary LiDAR modalities in the source dataset, while $\mathcal{L}_{\mathrm{DA}}$ pertains to the domain discrimination loss observed between the auxiliary target HS data and the base source HS data.

At this point, mimicking (2) and (3), we perform category prediction for the query data $(x_{\mathrm{q}}, y_{\mathrm{q}}) \in Q$ in the task and calculate the cross-entropy classification loss

$$p_\Theta\left(y_{\mathrm{q}} = c \middle| x_{\mathrm{q}}\right) = \frac{\exp\left(-d\left(\mathcal{F}_\theta\left(x_{\mathrm{q}}\right), \mathrm{P}_c^{(2)}\right)\right)}{\sum_{c=1}^{N} \exp\left(-d\left(\mathcal{F}_\theta\left(x_{\mathrm{q}}\right), \mathrm{P}_c^{(2)}\right)\right)} \tag{19}$$

$$\mathcal{L}_{\mathrm{fsl}} = \mathbb{E}_{S,A_1,A_2,Q}\left[-\sum_{(x_{\mathrm{q}},y_{\mathrm{q}})\in Q} \log p_\Theta\left(y_{\mathrm{q}} = c \middle| x_{\mathrm{q}}\right)\right] \tag{20}$$

where $\Theta = \{\theta, \varphi, \phi, \vartheta, \psi\}$ represents the set of learnable parameters of the model.

E. Training and Testing

1) Training: The total loss of the model for a task can be denoted as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{fsl}} + \mathcal{L}_d + \eta \times \left(\mathcal{L}_c^{A_1} + \mathcal{L}_c^{A_2}\right) \tag{21}$$

where η represents the weight factor of the auxiliary classification loss. Algorithm 1 displays the training procedure of I²MEP-Net.

2) Testing: In the testing process of the model, the target few-shot labeled data are encoded by the trained prototype network and then learn to obtain an NN Classifier to predict the category of the target unlabeled data $\mathsf{U} \subset \mathsf{X}_{\mathrm{novel}}^{\mathcal{T}}$

$$y = \arg\max_c \sum_{x_i \in \mathcal{N}(x)} I\left(y_i = c\right) \tag{22}$$

where y is the predicted category of x, $\{x_i, y_i\} \in \mathsf{X}^{A_2}$, $\mathcal{N}(x)$ denotes the neighborhood of x, and $x \in \mathsf{U}$. I is the indicator function, and I is 1 when $y_i = c$ and 0 otherwise.

Algorithm 1 Training Procedure of I²MEP-Net

Input: X, X^{A_1}, X^{A_2}, $\mathcal{F}_\theta$, $\mathcal{G}_\varphi$, $\mathcal{H}_\phi$, $\mathcal{T}_\vartheta$, $\mathcal{D}_\psi$.
Output: The well-trained $\mathcal{F}_{\theta^*}$.

```
Initialize network parameters;
for i = 1 to Iterations do
    Randomly sampling and constructing meta-tasks
      {S, Q ⊂ X, A1 ⊂ X^{A1}, A2 ⊂ X^{A2}};
    Extract features and obtain initial prototype P_c^(0) by
      Eq.(6);
    // Inter-MEPL;
    Calculate the auxiliary prototype P_c^{A1};
    Obtain the enhanced prototype P_c^(1) by Eq.(12);
    // Intra-MEPL;
    Calculate the diversity generating features G_{c,i}^{A2};
    Obtain the enhanced prototype P_c^(2) by Eq.(17);
    Calculate the total loss L_total of the network by
      Eq.(21) and update the network parameters;
end
```

IV. Experiments

A. Experimental Datasets

The arrangement of the prototypical network-based few-shot learning necessitates that the number of categories in the source dataset should be equal to or more than that in the target domain. As shown in Fig. 4, in our experiment, we use the Houston 2013 dataset, which includes paired HS and LiDAR data, as the source dataset, while four unimodal HS datasets serve as the target ones, namely, the University of Pavia (UP), Tea Farm (TF), WHU-Hi-HanChuan (WHU), and MUUFL Gulfport (MUUFL). Details, such as the categories of the datasets, are listed in Table I.

1) Houston 2013: The Houston 2013 data are collected by the National Center for Airborne Laser Mapping (NCALM) around the University of Houston campus and surrounding areas in June 2012, sponsored by the American National Science Foundation. It is provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest[1]. This dataset consists of HS and LiDAR data sources, with HS encompassing 144 bands from a spectral range of 364–1046 nm, and has a size of 349 × 1905 pixels. The land cover is marked with 15 categories.

2) UP: The UP data are acquired by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor in Pavia University, Italy. The dataset has a size of 610 × 340 pixels and includes 103 spectral bands. The land cover is marked with nine categories.

3) TF: The TF data are collected by the Aerosol Pushbroom HS Imager (PHI) at the Fanglu tea farm, Changzhou, China.[2] The dataset has a size of 348 × 512 pixels and includes 80 bands from a spectral range of 417–855 nm. The land cover is marked with ten categories.

4) WHU: The WHU dataset is obtained by cropping a region from the WHU-Hi-HanChuan dataset[3] acquired on June

[1] https://hyperspectral.ee.uh.edu/?page_id=459
[2] https://doi.org/10.3974/geodb.2017.03.04.V1
[3] http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE I
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN SOURCE AND TARGET DATASETS

Class No. \ Datasets	Houston 2013		UP		TF		WHU		MUUFL	
	Class Name	Numbers	Class Name	Numbers	Class Name	Numbers	Class Name	Numbers	Class Name	Numbers
1	Healthy grass	1251	Asphalt	6631	Masson pine	5806	Strawberry	15040	Trees	23246
2	Stressed grass	1254	Meadows	18649	Bamboo forest	2318	Cowpea	9913	Mostly grass	4270
3	Synthetic grass	697	Gravel	2099	Tea plant	28428	Soybean	805	Mixed ground surface	6882
4	Trees	1244	Trees	3064	Reed	214	Greens	3166	Dirt and sand	1826
5	Soil	1242	Sheets	1345	Paddy	6809	Trees	17679	Road	6687
6	Water	325	Bare soil	5029	Sweet potato	817	Grass	270	Water	466
7	Residential	1268	Bitumen	1330	Caraway	429	Red roof	3715	Building shadow	2233
8	Commercial	1244	Bricks	3682	Weed	1861	Gray roof	12425	Buildings	6240
9	Road	1252	Shadow	947	Water body	6141	Bare soil	4221	Sidewalk	1385
10	Highway	1227	–	–	Building/Road	911	Road	5701	Yellow curb	183
11	Railway	1235	–	–	–	–	Bright object	65	Cloth panels	269
12	Parking lot 1	1233	–	–	–	–	Water	2645	–	–
13	Parking lot 2	469	–	–	–	–	–	–	–	–
14	Tennis court	428	–	–	–	–	–	–	–	–
15	Running track	660	–	–	–	–	–	–	–	–
Total	–	15029	–	42776	–	53734	–	75645	–	53687

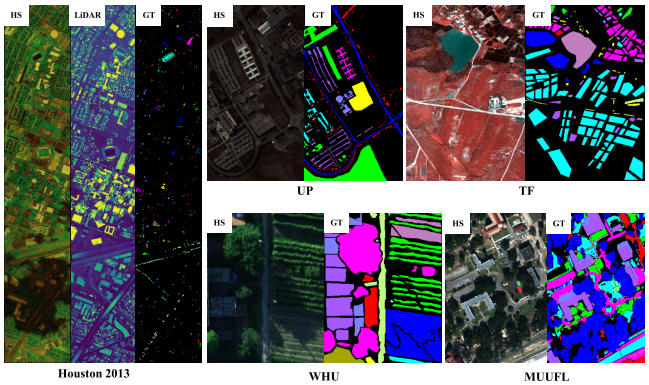


Fig. 4. Visualization of the public RS benchmark datasets: Houston 2013, UP, TF, WHU, and MUUFL.

17, 2016, in Hanchuan, Hubei, China. Specifically, the region cropped horizontally from the original spatial resolution of 1217 × 303 pixels starting at the 149th pixel and ending at the 534th pixel. The cropped region has a size of 386 × 303 pixels and includes 274 bands from a spectral range of 400–1000 nm. The land cover is marked with 12 categories.

5) MUUFL: The MUUFL data are collected in November 2010 at the Gulf Park campus of the University of Southern Mississippi, Long Beach, MS. This dataset has a size of 325 × 220 pixels and includes 64 bands from a spectral range of 375–1050 nm. The land cover is marked with 11 categories.

B. Experimental Setting

To validate the effectiveness of the proposed method, we compare it with several groups of state-of-the-art methods. These include classic machine learning method (SVM), popular deep learning methods (3D-CNN [34], HybridSN [35], SFormer [36], and SPFormer [37]), and state-of-the-art prototype learning methods (DFSL [38], DCFSL [39], Gia-FSL [40], CMFSL [41], and RPCL-FSL [42]). Among the SVM, 3D-CNN, HybridSN, SFormer, and SPFormer methods, in order to test the model's ability to handle a small number of labeled samples, we only use five labeled samples from per class in the target datasets for training, with the remainder used for testing purposes. The DFSL, DCFSL, Gia-FSL, CMFSL,

TABLE II
EXPERIMENTAL NETWORK SETTING

$\mathcal{F}_\theta$		$\mathcal{G}_\varphi$		$\mathcal{T}_\vartheta$	
Conv0	1,1,100	Conv1	3,3,8	Linear1	1,1,1024
Conv1	3,3,3,8	Conv2	3,3,8	Linear2	1,1,1024
Conv2	3,3,3,8	Conv3	3,3,8	Linear3	1,1,1024
Conv3	3,3,3,8	Max pooling1	2,2	Linear4	1,1,160
Max pooling1	2,2,4	Conv4	3,3,16	$\mathcal{D}_\psi$	
Conv4	3,3,3,16	Conv5	3,3,16	Linear1	1,1,1024
Conv5	3,3,3,16	Conv6	3,3,16	Linear2	1,1,1024
Conv6	3,3,3,16	Max pooling2	2,2	Linear3	1,1,1024
Max pooling2	2,2,4	Conv7	3,3,32	Linear4	1,1,1024
Conv7	3,3,3,32	$\mathcal{H}_\phi$		Linear5	1,1,1
		Conv1	1,1,160		

and the proposed methods all employ the same baseline network framework and identical experimental setups, such as size of input image patches $s = 9$, the number of HS channel mappings $b = 100$, the learning rate $lr = 0.001$, Adam optimizer, $N-$way $1-$shot $19-$query task settings, and training data made up of 300 randomly sampled labeled samples from each class in the source dataset and five labeled samples from each class in the target dataset. The RPCL-FSL method sets the task to $N-$way $2-$shot $19-$query due to the needs of the model. Furthermore, all hyperparameters for the comparative methods are set with publicly available values. The details of the proposed network are listed in Table II. Finally, the test results are determined by the average performance after ten iterations of each experiment. Model performance is evaluated using three metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficients (Kappa).

C. Hyperparameter Analysis

In this section, we will discuss the impact of several significant hyperparameters on the proposed model. On the one hand, the Inter-MEPL uses auxiliary LiDAR modality to generate additional prototypes to alleviate the instability of base prototypes under sparse samples. Among them, the enhanced prototypes are largely affected by the hyperparameter λ in (12), which reflects the extent of the role of the auxiliary prototypes. On the other hand, another vital hyperparameter in the proposed method is the weight factor

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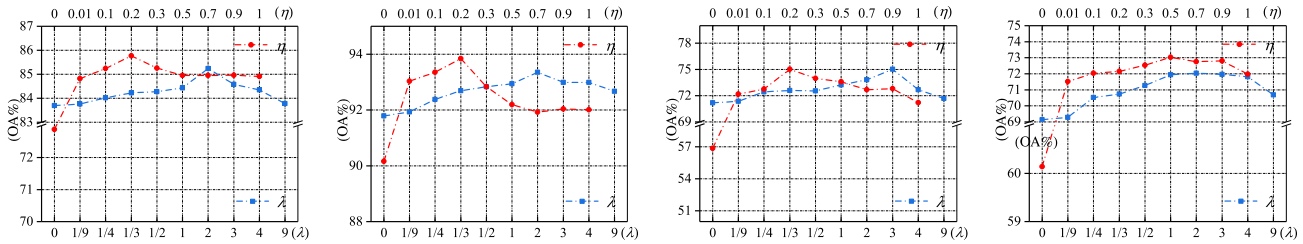


Fig. 5. AA of the four datasets under different hyperparameters λ and η. (a) UP. (b) TF. (c) WHU. (d) MUUFL.

TABLE III

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON THE TASK HOUSTON 2013→UP (FIVE LABELED SAMPLES PER CLASS)

Class No.	SVM	3D-CNN [34]	HybridSN [35]	SFormer [36]	SPFormer [37]	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
1	64.78±8.32	59.25±10.9	66.69±7.02	44.73±11.5	**87.16±3.54**	69.79±12.5	78.78±9.29	77.66±15.3	75.35±8.40	86.84±6.95	84.87±6.13
2	63.05±16.9	71.54±9.49	80.51±4.92	76.78±7.66	75.55±10.5	72.29±8.80	83.13±12.0	85.49±6.75	81.87±10.1	78.37±12.7	**87.56±8.29**
3	48.91±12.3	65.00±12.3	44.75±14.1	51.90±9.90	**85.07±8.17**	48.02±13.4	63.04±10.7	63.94±7.26	64.05±11.4	65.69±14.2	73.92±10.4
4	80.90±11.4	74.68±6.32	82.41±9.34	81.60±3.00	93.08±4.05	90.46±4.41	94.24±2.74	92.53±3.59	**96.23±1.98**	93.36±3.21	89.12±4.66
5	98.68±1.05	86.13±8.31	95.51±4.94	94.72±4.97	**100.0±0.00**	92.84±8.04	98.92±0.93	98.63±1.24	99.25±0.73	99.35±0.72	99.16±1.01
6	54.06±12.9	39.32±7.05	31.24±7.52	35.43±6.72	**92.21±8.43**	59.53±9.27	73.01±8.83	67.00±10.2	75.58±8.95	74.59±13.3	81.07±9.27
7	69.11±17.7	83.40±12.1	78.20±8.28	73.46±10.4	**99.55±0.74**	74.79±9.97	84.54±9.33	84.93±5.51	83.53±7.54	80.95±10.9	90.76±4.91
8	68.86±13.2	65.78±12.2	58.17±8.29	45.71±11.8	**90.46±5.18**	56.96±11.2	76.07±10.6	74.34±11.3	74.33±11.2	82.59±10.6	78.50±9.10
9	**99.73±0.26**	92.42±6.56	93.51±12.3	94.79±6.23	95.48±5.38	97.79±2.51	98.63±1.86	96.40±3.84	98.60±1.89	96.67±6.25	99.42±0.69
OA	65.46±5.75	66.54±4.48	69.72±2.59	64.26±2.90	84.26±4.02	70.48±3.57	81.35±4.66	81.23±3.39	80.59±3.85	81.19±4.85	**85.77±4.10**
AA	72.01±2.15	70.83±3.12	70.11±2.72	66.57±1.28	**90.95±1.88**	73.61±2.20	83.37±2.16	82.32±1.57	83.20±1.19	84.27±4.25	87.15±3.13
Kappa	56.70±5.73	57.72±4.73	60.56±2.96	54.09±3.02	80.18±4.66	62.62±3.89	76.02±5.29	75.64±3.94	75.13±4.28	75.97±5.62	**81.53±5.01**

TABLE IV

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON THE TASK HOUSTON 2013→TF (FIVE LABELED SAMPLES PER CLASS)

Class No.	SVM	3D-CNN [34]	HybridSN [35]	SFormer [36]	SPFormer [37]	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
1	67.72±12.8	75.81±9.81	85.22±9.49	87.36±4.38	93.28±6.56	89.76±5.86	92.09±4.34	91.22±3.40	91.99±3.12	89.25±9.17	**95.78±2.91**
2	42.52±7.20	48.92±14.6	32.36±13.8	37.60±11.5	**89.47±5.77**	60.64±8.05	78.69±5.32	79.52±7.03	76.04±6.03	82.56±9.89	80.46±4.63
3	75.12±7.77	74.95±18.9	91.04±2.66	86.89±9.23	81.69±9.32	87.47±3.10	92.09±2.74	91.74±3.03	91.18±3.86	90.08±5.52	**93.47±3.13**
4	88.28±8.34	99.28±1.01	88.71±27.8	99.52±0.74	**100.0±0.00**	99.76±0.24	**100.0±0.00**	99.90±0.19	99.86±0.43	99.33±0.86	**100.0±0.00**
5	94.18±5.21	95.96±4.04	97.41±1.84	97.08±1.20	97.50±4.36	95.65±3.21	99.26±0.46	98.91±0.69	98.60±1.97	96.89±4.21	**99.55±0.55**
6	92.17±5.08	87.77±6.72	80.32±13.8	82.67±11.7	88.77±5.45	84.17±11.5	95.22±4.62	95.06±2.67	**98.03±1.41**	93.67±6.24	88.85±7.39
7	90.35±4.33	82.41±18.5	72.31±17.5	78.82±8.61	**99.83±0.50**	90.99±7.40	99.27±0.78	94.60±5.82	94.06±9.56	96.86±4.11	97.85±1.73
8	46.41±8.40	41.45±7.34	24.98±17.0	36.94±9.30	**75.32±9.38**	46.35±10.4	64.05±10.2	62.16±10.1	60.27±18.2	67.19±15.1	70.25±11.7
9	98.86±0.29	98.41±0.97	**99.14±0.47**	99.12±0.43	98.94±0.35	99.10±0.40	98.95±0.31	98.81±0.64	99.04±0.42	99.03±0.40	99.12±0.36
10	91.58±6.70	71.56±17.1	82.21±10.5	91.60±6.80	93.81±6.63	92.92±4.29	98.25±2.71	98.34±1.43	**99.19±0.85**	97.70±3.11	97.68±2.47
OA	77.76±4.04	78.40±10.3	86.86±1.67	85.78±5.01	87.56±4.75	87.62±1.85	92.47±1.58	92.07±1.69	91.68±2.69	91.03±3.49	**93.84±1.82**
AA	78.72±1.70	77.65±3.21	75.37±4.63	79.76±1.91	91.86±1.05	84.68±2.33	91.79±1.13	91.03±1.72	90.83±3.43	91.26±3.61	**92.30±1.16**
Kappa	69.97±4.66	71.51±10.7	80.96±2.28	80.03±6.02	82.98±5.90	82.45±2.46	89.18±2.15	88.61±2.28	88.09±3.71	87.26±4.70	**91.08±2.50**

TABLE V

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON THE TASK HOUSTON 2013→WHU (FIVE LABELED SAMPLES PER CLASS)

Class No.	SVM	3D-CNN [34]	HybridSN [35]	SFormer [36]	SPFormer [37]	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
1	69.73±10.8	63.98±17.1	69.46±9.09	61.14±12.7	**91.23±4.66**	76.28±8.95	90.52±5.48	82.28±8.47	87.79±8.17	86.76±6.71	90.20±6.44
2	45.12±16.8	36.41±13.1	40.92±17.0	41.11±6.35	61.09±20.8	43.87±6.68	54.51±13.7	57.13±9.75	66.36±6.30	66.76±12.2	**70.16±8.03**
3	53.06±21.4	59.81±17.5	62.79±14.7	61.48±13.9	88.66±17.2	77.19±10.5	85.93±9.36	84.94±6.79	89.18±7.68	**94.61±7.86**	92.15±7.93
4	60.09±12.8	59.68±17.5	66.17±9.83	45.21±22.6	76.00±17.6	51.41±8.85	82.30±10.9	82.64±12.2	84.68±9.82	80.83±9.13	**87.98±3.87**
5	45.73±13.3	45.66±15.5	51.66±7.52	52.46±14.3	49.38±10.8	49.60±11.3	58.64±11.4	59.72±12.0	60.93±13.0	53.79±11.2	**61.12±7.03**
6	83.51±10.1	78.68±11.6	92.98±8.80	72.42±24.4	99.21±2.38	83.96±7.02	97.85±2.90	92.68±5.58	98.91±2.17	96.38±6.39	**99.25±1.21**
7	44.18±13.2	65.41±13.6	64.15±16.0	52.32±19.2	**75.48±11.6**	55.88±13.6	68.51±10.1	67.39±10.4	70.12±13.0	71.36±11.9	73.35±13.4
8	46.50±17.4	40.15±14.0	41.49±10.0	38.03±14.9	**74.61±17.6**	38.13±4.32	67.65±11.1	63.28±11.0	59.37±11.0	74.51±6.14	70.88±10.9
9	58.97±10.6	39.83±11.3	30.48±18.6	49.40±14.3	**80.91±8.68**	64.18±7.33	70.28±10.7	66.12±12.6	69.27±12.6	78.72±8.70	76.53±6.57
10	67.96±8.52	68.71±9.28	70.88±7.92	61.04±10.7	75.61±8.72	60.18±10.8	72.23±8.09	73.25±7.95	74.81±10.5	**76.04±9.33**	75.53±8.38
11	76.17±17.9	52.83±15.6	65.50±19.7	69.33±13.6	91.83±18.0	75.50±17.3	73.67±6.90	88.33±14.6	98.17±3.20	**98.83±3.50**	96.33±10.5
12	79.54±24.7	78.94±16.4	94.81±3.89	81.73±7.64	**96.58±3.15**	83.25±6.20	91.89±6.60	92.60±2.56	94.78±3.44	92.27±4.01	94.72±4.68
OA	54.90±5.34	51.59±9.32	55.39±4.73	51.69±5.54	71.79±4.17	55.87±3.48	70.68±3.86	68.72±2.77	71.33±3.47	72.48±4.23	**75.03±3.32**
AA	60.88±2.53	57.51±8.16	62.61±3.93	57.14±3.07	80.05±3.13	63.29±2.19	77.83±3.38	75.86±2.82	79.53±2.46	80.91±2.98	**82.35±1.87**
Kappa	49.04±5.55	45.72±9.63	49.76±5.08	45.43±5.70	67.76±4.62	50.13±3.38	66.36±4.27	64.17±3.13	67.14±3.73	68.47±4.67	**71.27±3.74**

η of the auxiliary classification loss in (21), indicating the significance of the auxiliary classification loss to the model. To this end, we design two sets of parameter analysis experiments, and Fig. 5 demonstrates the effects of different hyperparameter settings on the model's performance.

1) Selection of Hyperparameter λ: Experiments reveal that as the degree of the role of the auxiliary prototypes increases, the test accuracy is improved, but when the auxiliary prototypes are absolutely dominant, the ability of the base prototypes is ignored, and the test results turn out to be reduced. Moreover, comparing the optimal value with the scenario where the value is set to 0, it can be confirmed that the contribution of the auxiliary prototype to improving model performance is significant. Based on experimental results, the values of λ for the four datasets are set to 2, 2, 3, and 2, respectively.

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TABLE VI

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON THE TASK HOUSTON 2013→MUUFL (FIVE LABELED SAMPLES PER CLASS)

Class No.	SVM	3D-CNN [34]	HybridSN [35]	SFormer [36]	SPFormer [37]	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
1	65.96±11.6	60.51±9.25	71.08±8.26	58.85±8.13	**86.63±5.13**	66.11±5.96	75.96±6.58	70.17±8.02	74.31±6.60	79.51±3.33	79.94±4.42
2	**77.84±8.98**	59.99±14.6	53.73±18.3	55.75±15.6	69.59±16.7	54.81±8.34	63.17±13.1	62.47±10.2	63.57±13.5	60.10±11.8	69.79±9.46
3	54.50±12.6	46.76±6.94	28.48±12.2	27.84±5.97	37.91±16.4	33.06±7.84	46.01±10.1	42.19±10.2	48.53±11.2	38.97±11.9	**55.94±8.75**
4	73.22±9.70	60.60±10.4	62.24±17.8	64.76±13.7	**83.77±16.5**	63.10±14.8	72.04±8.37	63.83±10.7	65.76±15.9	63.99±9.22	77.36±12.7
5	58.14±27.9	57.55±14.4	57.30±13.6	52.86±14.3	56.60±30.5	48.76±10.7	63.39±8.22	58.79±7.23	58.77±6.86	57.59±9.86	**72.71±5.67**
6	79.33±24.8	71.67±10.6	44.97±20.5	47.22±16.2	**95.16±13.7**	62.58±14.4	70.39±16.6	70.76±19.4	75.05±15.1	87.09±14.2	88.26±12.9
7	50.28±23.8	62.06±9.55	49.50±16.0	51.83±15.8	80.53±14.2	52.93±10.6	72.07±7.99	67.61±11.3	75.47±9.35	76.52±6.55	**82.59±4.29**
8	47.28±16.6	34.13±10.2	23.48±11.3	30.52±12.6	**79.88±9.70**	42.26±6.67	52.14±8.36	49.63±9.43	54.19±7.12	68.98±8.97	69.15±5.18
9	43.35±7.76	32.73±9.32	41.99±8.69	30.56±5.46	29.44±17.4	41.25±8.31	49.41±12.7	48.33±10.1	**53.64±9.64**	49.57±7.96	42.67±9.58
10	62.30±14.8	43.15±12.8	57.25±10.8	48.20±10.3	46.18±29.4	52.70±8.68	62.58±9.82	57.64±9.97	**65.06±9.37**	64.04±9.54	64.33±11.4
11	85.38±9.89	50.38±8.99	74.66±19.0	86.86±9.01	**94.43±5.58**	84.70±5.19	90.34±6.08	91.63±6.10	91.36±7.56	93.86±5.53	89.24±11.3
OA	61.50±6.13	54.61±4.35	54.79±2.08	49.77±3.05	72.66±4.15	54.77±3.36	65.77±3.43	61.35±3.97	65.16±2.50	67.48±2.75	**73.03±3.14**
AA	63.42±3.41	52.68±1.85	51.33±3.59	50.48±2.62	69.10±2.96	54.75±2.35	65.23±2.64	62.10±2.63	65.97±2.54	67.29±2.25	**72.00±3.13**
Kappa	52.81±6.56	44.86±4.40	43.77±2.18	39.32±2.90	65.18±4.84	44.52±3.69	57.20±3.81	52.38±4.34	56.69±2.73	59.20±3.23	**66.00±3.75**

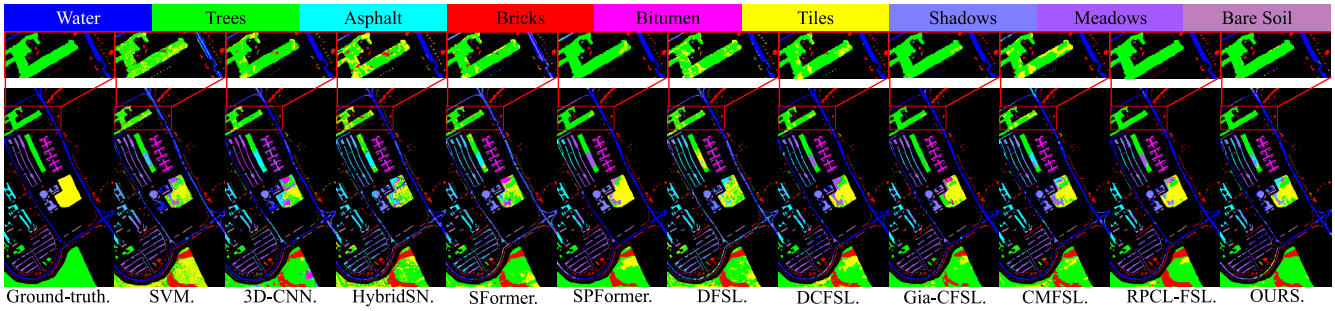


Fig. 6. Classification maps for different classifiers in UP.

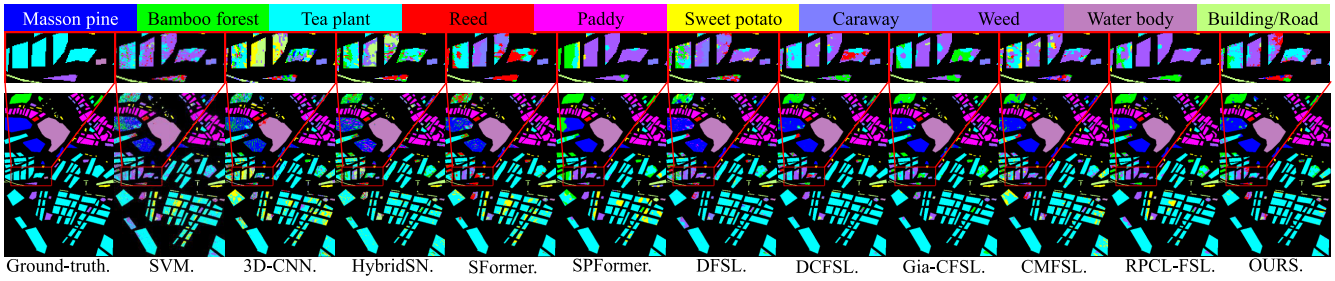


Fig. 7. Classification maps for different classifiers in TF.

2) Selection of Hyperparameter η: The lowest classification accuracy is achieved when this parameter is set to 0, which laterally reflects that the auxiliary classification loss has a positive effect on the model. Moreover, from the observation of experimental results, it can be found that the optimal classification accuracy of the model does not occur when the parameter is set to 1. At this time, the auxiliary classification loss and the main loss are of equal weight, which increases the model's attention to the auxiliary task that might interfere with its handling ability of the primary task. According to the experimental results, different values are chosen for the four datasets in the experiment, i.e., $\eta = 0.2, 0.2, 0.2$, and 0.5.

Significantly, our observations from the experimental results indicate that most of the parameter setting values exhibit insensitivity to changes in the model's performance. This phenomenon predominantly occurs near the optimal values of parameter λ, leading us to analyze that within this range of values, the inclusion of the auxiliary prototype continues to exert a positive influence on the model's performance. In addition, both types of prototypes may provide similar or overlapping information, thus maintaining the model's performance relatively stable within a certain range.

D. Comparison With State-of the-Art Methods

The test results on four public datasets are shown in Tables III–VI, where the optimal results are boldly emphasized and the suboptimal results are underlined. We analyze and summarize the performance of different methods.

1) *Proposed Method Versus Nonprototype Learning Methods:* SVM is somewhat capable of suppressing overfitting issues and demonstrates a certain robustness against noisy data, maintaining relatively stable classification performance across four datasets. The methods, such as 3D-CNN, HybridSN, and SFormer, leverage deep learning to mine features from high-dimensional data, training the classification model through extensive data. However, experiments reveal that limited labeled data can result in a risk of overfitting, which significantly impacts the performance of these methods, especially within the SFormer method where the problem of data hungry is even more pronounced. As an advancement for few-shot problems, the SPFormer method retains its substantial feature extraction capabilities while streamlining the model parameters to enhance its ability to handle small samples. Observations indicate that this method exhibits performance close to the proposed

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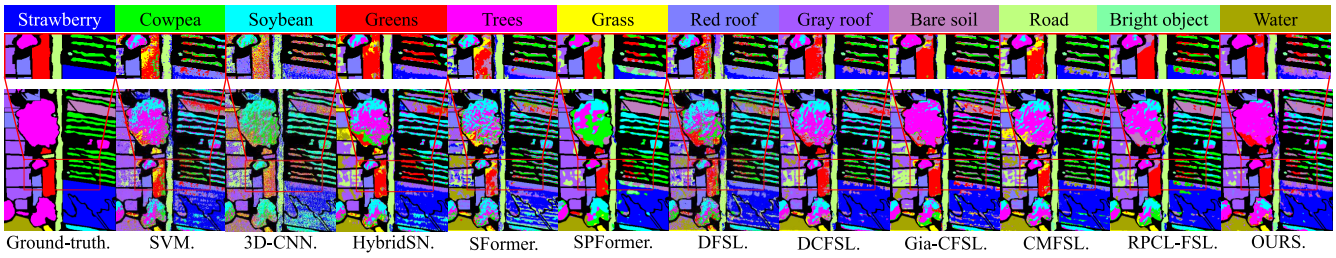


Fig. 8. Classification maps for different classifiers in WHU.

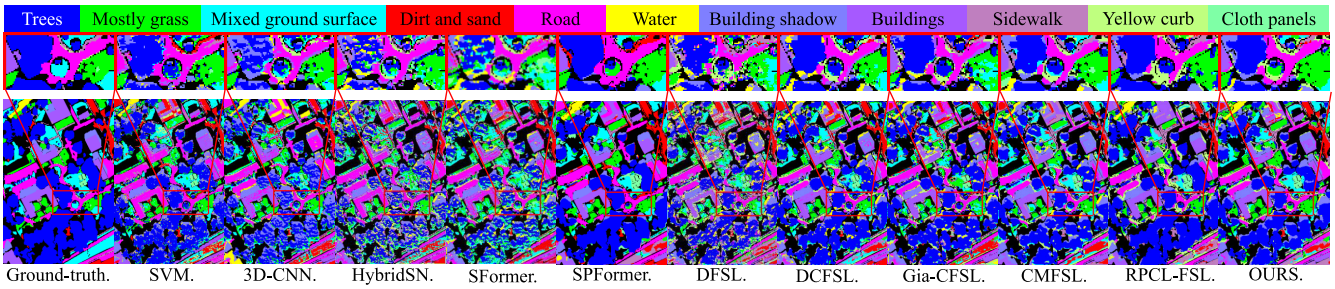


Fig. 9. Classification maps for different classifiers in MUUFL.

method on most datasets but is highly susceptible to class imbalance. For instance, although the AA of the SPFormer method is slightly higher than that of the proposed method on the UP dataset, the OA is reduced due to the prevalence of minority classes. Meanwhile, the proposed method consistently demonstrates commendable performance on a wider range of datasets.

2) *Proposed Method Versus Prototype Learning Methods:* Methods, such as DFSL, DCFSL, Gia-CFSL, CMFSL, and RPCL-FSL, are all categorized under prototype learning, almost covering the currently publicly available state-of-the-art prototype-based HSIC technologies. DFSL initially introduces a novel approach to address few-shot HSIC challenges by learning a prototype metric space; DCFSL and Gia-CFSL further reduce the impact of data distribution discrepancies on prototype learning through adversarial domain adaptation and graph information aggregation; CMFSL improves distance calculations in the metric space while addressing data distribution differences, and RPCL-FSL is designed with triple constraints to stabilize and refine prototypes concerning their instability. Experiments show that the DFSL method displays a degree of few-shot handling capability. DCFSL, Gia-CFSL, and CMFSL either improve upon data distribution differences or optimize measurement methods, resulting in enhanced classification accuracy. Nevertheless, they still face risks associated with unstable prototype representations under sparse samples. Especially, in complex scenarios with challenging classifications, such as the WHU and MUUFL datasets, the RPCL-FSL method, that refines base prototypes using additional prototypes derived from query sets, consistently outperforms them. In contrast, the proposed method leverages additional modal information to enhance the stability of prototype representation, showcasing higher classification performance across all four datasets. This phenomenon reaffirms the significant role of supplementary information in stabilizing prototype representation.

Figs. 6–9 present the visual classification results of different methods, highlighting specific areas for comparison. Corresponding to the aforementioned analytical conclusions, the classification maps of SVM, 3D-CNN, HybridSN, SFormer, and DFSL methods all exhibit noticeable noise and many misclassifications. The SPFormer method, on the other hand, shows clear differences in category classification in the classification maps, e.g., the highlighted region in Fig. 9 is closer to the true image, while the zoomed-in region in Fig. 7 appears to be almost completely misclassified. The overall classification maps of other prototype methods show improvements, with the proposed method performing the best, followed by the RPCL-FSL method.

V. Discussion

A. Ablation Study

The proposed I²MEP-Net innovatively introduces the Inter-MEPL and Intra-MEPL to obtain stable representative prototypes. Therefore, we designed an ablation study, as shown in Table VII, to verify the function of each module in the proposed method. In the table, pn represents the baseline network, w/o Inter-MEPL indicates the removal of Inter-MEPL process from the proposed method, and w/o Intra-MEPL signifies the removal of Intra-MEPL process. An analysis of the ablation experiment reveals that regardless of which process is removed, its test accuracy decreases compared with our proposed method, but still shows a certain improvement over the baseline network. Among them, the results of w/o Intra-MEPL show a larger decrease compared with OURS. We believe that this outcome largely attributes to the positive contribution of target information in Intra-MEPL toward model learning and generalization abilities.

To further demonstrate the performance of the proposed enhancing prototype learning, we proceed to visualize the test results under different prototype representations. Stable representative prototypes in the proposed method are provided

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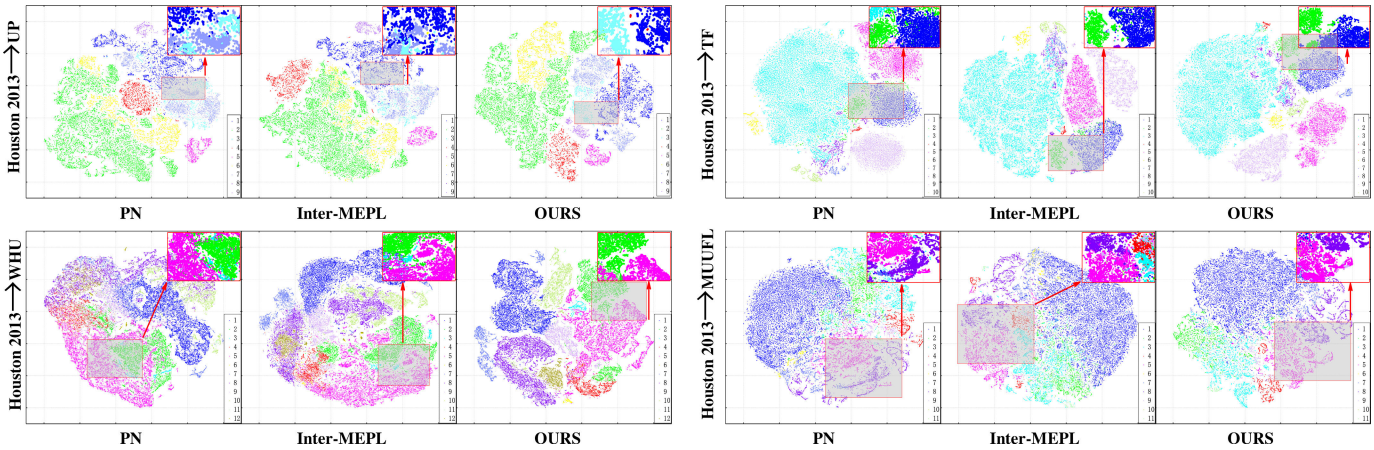


Fig. 10. Visualization of class separability on the target datasets.

TABLE VII
ABLATION EXPERIMENTS RESULTS (%) ON THE TARGET DATASETS

Dataset		PN	w/o Inter-MEPL	w/o Intra-MEPL	OURS
UP	OA	70.48±3.57	83.69±4.72	74.32±3.14	**85.77±4.10**
	AA	73.61±2.20	84.80±2.12	75.11±1.39	**87.15±3.13**
	Kappa	62.62±3.89	79.00±5.45	66.79±3.70	**81.53±5.01**
TF	OA	87.62±1.85	91.79±1.96	90.61±1.49	**93.84±1.82**
	AA	84.68±2.33	90.60±1.41	86.60±2.21	**92.30±1.16**
	Kappa	82.45±2.46	88.94±2.64	86.41±2.09	**91.08±2.50**
WHU	OA	55.87±3.48	68.73±4.14	61.73±3.75	**75.03±3.32**
	AA	63.29±2.19	77.26±3.16	66.37±2.35	**82.35±1.87**
	Kappa	50.13±3.38	64.30±4.47	54.11±3.85	**71.27±3.74**
MUUFL	OA	54.77±3.36	69.13±3.02	60.94±2.77	**73.03±3.14**
	AA	54.75±2.35	68.02±2.58	57.70±2.31	**72.00±3.13**
	Kappa	44.52±3.69	61.31±3.20	51.30±2.90	**66.00±3.75**

by the HS feature encoder; hence, we utilize the T-SNE method to visualize HS-encoded features, demonstrating the ability of prototype representations. Fig. 10 shows the feature visualization results on four target datasets, where pn represents the base prototype, Inter-MEPL denotes the enhanced prototype after Inter-MEPL, and OURS stands for the final enhanced prototype after Inter-MEPL and Intra-MEPL. Comparison between the test results under different enhanced prototypes and pn clearly depicts that the capacity of prototype representations has been improved after enhancing prototype learning. For instance, the separability between class "Asphalt" and "Bitumen" in the UP dataset, between class "Bamboo forest" and "Reed" in the TF dataset, between class "Cowpea" and "Trees" in the WHU dataset, and between class "Buildings" and "Mixed ground surface" in the MUUFL dataset has all gradually enhanced with enhancing prototype learning.

B. Extended Study of Cross-Modal Auxiliary Data

Based on the conclusion emphasized in the ablation experiment that the Inter-MEPL process has a positive impact on the proposed network, we attempt, in this section, to dissect the specific conditions and modes of operation of the cross-modal auxiliary data in the Inter-MEPL process in two ways. Specifically, the auxiliary modality injects richness into the learning of the base embedding space through prototype alignment and weighting. On the one hand, we believe that there does not need to be an absolute semantic correspondence between the auxiliary LiDAR data and the base HS data while controlling for class alignment. On the other hand, we posit that different types of auxiliary data, such as SAR modality, can still collaborate with the base modality in cross-modal learning and achieve the effect of prototype enhancement. Accordingly, we design two sets of experiments for verification.

1) Selection of Different Scenarios: We design the experiment shown in Table VIII. Based on the unimodal prototype learning (TF→UP), we borrow LiDAR data from Houston 2013 to provide additional meta-knowledge to adapt to our proposed I^{2}MEP-Net. The classes of the LiDAR data are repartitioned and mapped to target classes. Here, we define a mapping function $\Gamma : \mathsf{X}^{\mathcal{A}_1} \rightarrow \bar{\mathsf{X}}^{\mathcal{A}_1}$, where $\Gamma(x) = M[x-1]$ corresponds to the xth element of the mapping matrix. A class mapping matrix $M = [1, 1, 1, 2, 3, 4, 5, 5, 6, 7, 7, 8, 8, 9, 10]$ is given according to the number of classes in the TF dataset, after which the LiDAR data are repartitioned to get $\bar{\mathsf{X}}^{\mathcal{A}_1}$. At this point, HS data X from TF, LiDAR data $\bar{\mathsf{X}}^{\mathcal{A}_1}$ from Houston 2013, and HS data $\mathsf{X}^{\mathcal{A}_2}$ from UP are fed into the proposed I^{2}MEP-Net as inputs for Inter- and Intra-MEPL. The classification results reveals that even in the absence of pairings of multimodal data in the source dataset, the proposed method can still provide effective additional knowledge to enhance the stability of the base prototype under sparse samples. This is achieved by aligning classes from auxiliary data from different scenes and modalities. Such a finding helps to reduce the application constraints of the proposed method under special scenarios.

2) Selection of Different Modalities: We conduct the experiments, as shown in Table IX. HS-SAR Berlin→Augsburg indicates the use of a new HS-SAR Berlin dataset with eight categories as the source domain, in which SAR serves as an auxiliary modality to collaborate with the base HS modality for Inter- and Intra-MEPL on the target HS Augsburg dataset with seven categories. The classification results of different methods are demonstrated in the right column of the table. Experimental results indicate that when the cross-modal auxiliary data in the proposed method are switched from LiDAR to SAR, the proposed model still exhibits superior performance. Furthermore, we design the HS-LiDAR Houston 2013→Augsburg experiment, in which the proposed method still achieved the best performance. A comparison reveals that the results obtained from training with the source domain

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TABLE VIII
CLASSIFICATION ACCURACY (%) ON THE TASK TF→UP (FIVE LABELED SAMPLES PER CLASS)

Methods	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
OA	70.09±4.31	79.57±3.31	80.29±2.84	79.70±3.26	80.89±5.99	**84.09±4.96**
AA	73.23±1.22	80.52±1.36	80.61±2.21	82.42±1.94	85.29±2.22	**86.23±1.77**
Kappa	62.09±4.51	73.61±3.96	74.48±3.40	73.98±3.84	75.84±6.71	**79.58±5.74**

TABLE IX
CLASSIFICATION ACCURACY (%) ON THE TASK HS-LIDAR HOUSTON 2013→AUGSBURG AND THE TASK HS-SAR BERLIN→AUGSBURG (FIVE LABELED SAMPLES PER CLASS)

Methods		HS-LiDAR Houston 2013 →Augsburg	HS-SAR Berlin →Augsburg
DFSL [38]	OA	57.64±2.86	56.33±2.75
	AA	54.21±2.95	47.99±3.72
	Kappa	45.21±2.96	42.41±2.62
DCFSL [39]	OA	70.86±4.09	65.34±5.18
	AA	64.67±2.91	60.31±4.60
	Kappa	61.30±4.95	54.71±6.01
Gia-CFSL [40]	OA	66.38±5.17	60.72±4.49
	AA	60.99±3.89	58.14±4.39
	Kappa	56.18±6.02	48.95±5.24
CMFSL [41]	OA	74.19±4.09	70.47±5.61
	AA	67.25±1.70	65.61±3.34
	Kappa	65.75±4.87	61.39±6.62
RPCL-FSL [42]	OA	74.65±7.96	72.39±5.75
	AA	65.73±4.01	65.78±4.06
	Kappa	66.67±9.37	63.83±0.46
OURS	OA	**76.99±5.22**	**74.86±4.56**
	AA	**67.37±3.47**	**67.01±4.06**
	Kappa	**69.08±6.51**	**66.50±5.59**

dataset of Houston 2013 are generally improved across all methods compared to those trained with the Berlin dataset as the source domain. Since the two source domain datasets come from different scenarios, we can only roughly attribute this phenomenon to the richness of the categories in the source domain datasets having a significant impact on the generalization performance of the model. This explanation is also pointed out in the literature [39].

C. Analysis of Few-Shot Classification Capability

At present, there are two effective means to handle the few-shot classification problem. One is based on multimodal fusion technology, which utilizes additional information provided by complementary perspectives and modalities to enhance the class representation ability under sparse samples and improve classification accuracy. The other is based on knowledge transfer and domain adaptation, which uses learned effective knowledge for few-shot classification. Due to the high cost of obtaining data from different perspectives and modalities in practical applications, the latter has received more research and application. The method we propose is a development on the latter, proposing a multimodal knowledge transfer method to solve the unimodal few-shot classification problem. Here, we design the comparison experiments between the multimodal fusion methods and the proposed method. The aim of the experiments is to evaluate their few-shot classification capability for the publicly available paired HS and LiDAR data in the MUUFL dataset. In the experiments, five labeled samples of each class are used for training, while the remaining samples are used for testing.

Multimodal methods, such as CoupledCNNs [43], AsyFFNet [44], DFINet [45], CACL [46], and MFT [47], are state-of-the-art methods designed specifically for the fusion classification of HS and LiDAR data. These have been proven to outperform unimodal classification performance due to their effective information fusion capabilities. However, under training conditions with few-shot data, despite the presence and utilization of complementary LiDAR modality information for fusion classification, the accuracy of these multimodal fusion methods still falls short of our proposed method, as shown in Table X. Especially, on the class classification evaluation metric AA, our method demonstrates absolute competitive advantage, further verifying its sufficient ability to handle few-shot problems.

D. Sequential Discussion of Enhancing Prototype Learning

In our proposed method, we first use the Inter-MEPL to compensate for the sparsity of few-shot features using auxiliary LiDAR prototype and then enrich sample diversity representation based on the Intra-MEPL to address the loss of diversity in the previous step. However, these two enhancing prototype learning processes each have a certain positive effect on the stability of representative prototypes. In this section, we design a set of experiments to discuss the effect on the stability of representative prototypes, in terms of the sequence of these two enhancing prototype learning processes, as shown in Table XI.

Analyzing Table XI, we find that the sequence of the two enhancing prototype learning processes contributes to the classification results, and the sequence of Inter-MEPL and then Intra-MEPL is more conducive to the stable characterization of representative prototypes. We believe that the Intra-MEPL under sparse samples first is largely affected by the ability of class representation, while the class prototypes are more stable after the Inter-MEPL, and the Intra-MEPL at this time will have twice the effect with half the effort.

E. Running Efficiency Evaluation

The running times of different methods are shown in Table XII. Considering that the prototype-based comparative method uses the same source dataset for training, we only present the average training time to provide a more intuitive evaluation. It is observed that compared with nonprototype learning methods, prototype-based methods require more computational overhead. In particular, the proposed method utilizes additional multimodal auxiliary information for enhancing prototype learning, which inevitably incurs more additional computational overheads. However, once the training is completed, i.e., the prototype learning-based methods obtain stable representative prototypes, they have similar testing times on new classes. Therefore, we believe that the proposed method can still be positively evaluated in terms of running efficiency.

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TABLE X
FEW-SHOT CLASSIFICATION ACCURACY (%) ON THE MUUFL DATASET (FIVE LABELED SAMPLES PER CLASS)

Methods	CoupledCNNs [43]	AsyFFNet [44]	DFINet [45]	CACL [46]	MFT [47]	OURS
OA	69.25±5.02	67.73±5.60	70.46±2.54	71.90±3.50	70.43±2.71	**73.03±3.14**
AA	52.51±4.37	55.57±3.96	55.59±2.49	54.99±2.98	64.17±4.08	**72.00±3.13**
Kappa	61.25±5.91	59.54±6.31	62.47±2.97	64.20±4.22	62.39±3.15	**66.00±3.75**

TABLE XI
CLASSIFICATION ACCURACY (%) FOR DIFFERENT ENHANCING PROTOTYPE LEARNING SEQUENCES

Dataset		UP	TF	WHU	MUUFL
OURS ((Intra- and Inter-MEPL))	OA	82.81±4.21	91.63±2.23	71.59±4.03	71.50±2.56
	AA	84.41±3.73	88.58±3.17	80.12±2.92	70.90±2.88
	Kappa	77.75±5.07	87.95±3.04	67.57±4.44	64.14±2.97
OURS ((Inter- and Intra-MEPL))	OA	**85.77±4.10**	**93.84±1.82**	**75.03±3.32**	**73.03±3.14**
	AA	**87.15±3.13**	**92.30±1.16**	**82.35±1.87**	**72.00±3.13**
	Kappa	**81.53±5.01**	**91.08±2.50**	**71.27±3.74**	**66.00±3.75**

TABLE XII
RUNNING TIME OF DIFFERENT METHODS ON THE TARGET DATASETS

Dataset		SVM	3D-CNN [34]	HybridSN [35]	SFormer [36]	SPFormer [37]	DFSL [38]	DCFSL [39]	Gia-CFSL [40]	CMFSL [41]	RPCL-FSL [42]	OURS
UP	Testing time(s)	0.17	1.02	1.90	2.24	4.85	3.28	3.04	3.14	3.93	3.31	3.09
TF	Testing time(s)	0.22	1.26	2.02	2.58	4.08	4.00	3.71	3.77	4.74	3.73	3.77
WHU	Testing time(s)	0.42	2.22	1.86	2.49	4.89	3.49	3.56	3.51	4.07	3.74	3.97
MUUFL	Testing time(s)	0.23	1.22	1.48	2.66	4.02	3.76	3.70	3.86	4.77	3.69	3.69
Average training time(min)		–	0.32	0.35	5.31	5.99	12.3	19.6	35.1	18.8	4.96	17.3

VI. CONCLUSION

The method proposed in this article designs an Inter- and Intra-MEPL module to obtain stable representative prototypes, solving problems affecting prototype learning, such as insufficient training samples, spectral mixing, and noise interference in complex scenarios. The proposed I^2MEP-Net first introduces an auxiliary LiDAR modality for Inter-MEPL, which compensates for the sparsity of few-shot features in a cross-modal learning manner, and then designs an adaptive diversity feature generation method for Intra-MEPL to enhance individual characteristics of few-shot features and enrich the prototype embedding space. Specifically, our method makes a preliminary attempt to not require semantic relevance between the auxiliary modality and the base modality, which reduces the application constraints of the proposed model. Moreover, this work wants to express that methods based on multimodal prototype learning are no longer confined to visual and textual modalities, or paired dual modalities. Future work can focus more on available multimodal remote sensing data.

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Wenzhen Wang received the M.S. degree from the School of Electric and Electronic Engineering, Shanghai University of Engineering Science, Shanghai, China, in 2021. She is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

Her research interests include deep learning and remote sensing image processing.



Fang Liu (Member, IEEE) received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is an Associate Professor with Nanjing University of Science and Technology, Nanjing, China. She is with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, and also with Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology. Her research interests include deep learning, object detection, polarimetric SAR image classification, and change detection.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2014, he has been the Deputy Director of Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. He was the Second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science. He has published more than 200 international journal articles, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. His main research interests include inverse problems in image processing, computer vision and image understanding, pattern recognition, and remote sensing.

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